Using MLR to Predict the Transfer Value of English Premier League Players

By Shubham Tanna · A replication of Li (2020), "Multiple regression model for predicting soccer player value in English Primer League"

Introduction:

Football, also known as association soccer, is a global sport with a massive following. Soccer clubs now operate as businesses that make critical decisions on which players to employ. The transfer value of a player is of utmost importance to these companies and is influenced by various factors, such as physical condition, player performance, and club rank. This study aims to investigate the correlation between player value and variables from these three categories, using data from the English Premier League from 2018 to 2019. The results of this study could provide valuable insights for soccer clubs in selecting suitable players and offering reasonable revenue during transfers.

Method:

The independent variables are the variables assumed to affect the dependent variable, which is the player's transfer value (log-transformed). The independent variables chosen are age, height, weight, position, average playing time (APT1), average appearance time (APT2), average goals (AG), average assists (AS), average clean sheets (ACS), penalty goals (PG), penalty misses (PM), red cards (RC), yellow cards (YC), player rank (PR), and club rank (CR). The dependent variable is the transfer value, and the data is collected from the official premier league website, transfer market website and footy stats dot org website.

Hypothesis and Data:

Age, height, weight, position, APT1, APT2, AG, AS, ACS, PG, PM, RC, YC, PR, and CR are assumed to affect the transfer value of players. The data collected from the premier league season 18-19 includes 324 (Table 1) rows of players whose data is obtained.

Results:

The descriptive stats (Table 2) show the basic information about the data like minimum, maximum, average, etc., to help us gain a better picture of the data collected. The correlation analysis between each independent variable and the dependent variable is shown in the table. The values in the Table 3 are Pearson's coefficients (r). A simple and brief judgment about the relationship between player value and 15 independent variables is made through the correlation analysis. A single asterisk shows a p-value less than 0.05, double shows a p-value less than 0.01, and no asterisk shows a p-value greater than 0.05.

In the regression analysis (Table 4a, 4b, and 4c), the p-value is obtained by t-test. P value less than 0.05 means that the corresponding X has a significant influence on the dependent variable. We use these 15 predictors as the independent variables and player value is used as the dependent variable in the linear regression analysis. The R-squared value is 0.684, indicating that the model explains 68.4% of the variation in player value. The F-test of the model found that the model passed the F-test ($F=44.237$, $p=0.000<0.05$), which means that at least one of these 15 independent variables will influence Player value. The test for the model's multicollinearity found that all the VIF values in the model are less than 5, indicating no serious multicollinearity among the 15 explanatory variables. The D-W value is near 2, indicating no autocorrelation in the residuals.

Discussion:

According to the results, age, average playing time, average goals and club rank significantly impact the market value of soccer players, whereas height, weight, position, average assists, average clean sheets, penalty goals, penalty misses, red cards, yellow cards, and player rank do not have a significant impact on the player value, which contradicts our hypothesis. Average appearance time is borderline ($p = 0.056$). The present study is subject to certain limitations. Primarily, the scope of the analysis is restricted to data exclusively drawn from the 2018-2019 Premier League season. Consequently, the findings may solely delineate a few factors that affect the valuation of players during the said season. Additionally, it is worth noting that the model explains a modest 68.4% of the variation in player value ($R^2 = 0.684$, adjusted $R^2 = 0.668$). This could be attributable to the small sample size or the inclusion of certain independent variables that fail to sufficiently explain player valuation. Hence, if one intends to obtain more desirable prediction outcomes, it may be imperative to eliminate the superfluous variables.

Table 1: Data

	Player	Age	Height	Weight	Position	Average playing time	Average appearances time	Average goals	Average assists	Average clean sheets	Penalty goals	Penalty misses	Red cards	Yellow cards	Player rank	Club rank	Player value
0	Aaron Cresswell	3	2	1.0	4	6.677713	9.0	0.0	0.5	1.5	0	0	0	1	20	10	15.581952
1	Andriy Yarmolenko	3	3	3.0	2	5.435903	2.5	1.0	0.0	0.5	0	0	0	2	7	10	15.961442
2	Andy Carroll	3	4	2.0	2	5.424950	1.5	0.0	0.0	1.0	0	0	0	2	17	10	14.873301
3	Angelo Ogbonna	3	4	3.0	4	6.840547	10.0	0.5	0.5	1.5	0	0	0	2	13	10	15.581952
4	Arthur Masuaku	2	2	2.0	4	6.777647	9.5	0.0	0.5	3.0	0	0	0	2	25	10	15.581952
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
329	Matt Doherty	2	3	3.0	4	7.361058	17.5	2.0	2.5	4.5	0	0	0	5	5	7	16.482739
330	Morgan Gibbs-White	1	2	NaN	1	5.779199	2.5	0.0	0.5	2.5	0	0	0	1	16	7	15.684231
331	Romain Saiss	3	4	3.0	1	6.349139	6.0	1.0	0.0	2.0	0	0	0	7	7	7	15.789592
332	Ryan Bennett	3	3	2.0	4	7.333023	17.0	0.5	0.0	4.5	0	0	0	12	10	7	15.096444
333	Willy Boly	2	4	4.0	4	7.368024	18.0	2.0	0.0	4.0	0	0	1	3	4	7	16.384299

324 rows × 17 columns

Table 2: Descriptive Stats

Variable	Count	Mean	Standard Deviation	Minimum	25%	50%	75%	Maximum
Player Value	323.00	15.96	1.06	13.49	15.28	15.96	16.65	18.56
Age	323.00	2.10	0.70	1.00	2.00	2.00	3.00	3.00
Height	323.00	2.80	0.72	1.00	2.00	3.00	3.00	4.00
Weight	323.00	2.20	0.79	1.00	2.00	2.00	3.00	4.00
Position	323.00	2.36	1.32	1.00	1.00	2.00	4.00	4.00
APT1	323.00	6.34	1.15	0.00	5.99	6.71	7.13	7.44
APT2	323.00	9.10	5.79	0.00	4.00	9.00	14.50	19.00
AG	323.00	1.14	1.88	0.00	0.00	0.50	1.50	11.00
AS	323.00	0.87	1.28	0.00	0.00	0.50	1.00	7.50
AC	323.00	3.13	2.32	0.00	1.50	2.50	4.50	10.50
PG	323.00	0.18	0.76	0.00	0.00	0.00	0.00	7.00
PM	323.00	0.05	0.30	0.00	0.00	0.00	0.00	3.00
RC	323.00	0.10	0.30	0.00	0.00	0.00	0.00	1.00
YC	323.00	2.78	2.69	0.00	1.00	2.00	4.00	13.00
PR	323.00	12.58	7.42	1.00	6.00	12.00	18.50	29.00
CR	323.00	10.18	5.79	1.00	5.00	10.00	15.00	20.00

Table 3: Individual correlation of independent variables with the dependent variable.

Variable	r Value	Variable	r Value
Age	-0.349**	**AS**	0.416**
Height	-0.01	**AC**	0.577**
Weight	-0.053	**PG**	0.205**
Position	-0.052	**PM**	0.112*
APT1	0.389**	**RC**	0.045
APT2	0.403**	**YC**	0.105
AG	0.495**	**PR**	-0.358**
		CR	-0.509**

*p<0.05 **p<0.01

Table 4a, 4b, and 4c: MLR results.

Variable	T	P	VIF
Constant	45.725	0.000	-
Age	-12.319	0.000	1.092
Height	1.664	0.097	1.692
Weight	1.117	0.265	1.640
Position	-0.472	0.637	1.290
APT1	3.887	0.000	3.463
APT2	1.921	0.056	4.732
AG	4.915	0.000	3.229
AS	0.799	0.425	1.803
AC	1.745	0.080	3.076
PG	-1.427	0.155	2.152
PM	0.385	0.700	1.540
RC	-0.084	0.933	1.162
YC	-0.912	0.358	1.757
PR	0.361	0.718	2.206
CR	-9.927	0.000	1.647

F	44.237
R^2	0.684
Adjusted R^2	0.668
D-W	1.82

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Reference: Li, S. (2020). Multiple regression model for predicting soccer player value in English Primer League. The Frontiers of Society, Science and Technology, 2(15), 132–143. https://doi.org/10.25236/FSST.2020.021516